# An Empirical Study of a Vision Foundation Model for Remote Sensing Wind Turbine Detection

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***Abstract*—Vision Foundation Models (VFMs) pretrained via self-supervised learning exhibit strong semantic representation capabilities across diverse vision tasks. However, their effectiveness in remote sensing small object detection remains insufficiently understood. This paper presents an empirical study on wind turbine detection in high-resolution remote sensing imagery, where targets occupy approximately 1/40 of the image area. We integrate a Vision Transformer–based foundation model, DINOv3, into a unified YOLOv11 detection framework and conduct controlled comparisons under different input resolutions (640 and 1024) and feature integration strategies. Experimental results show that DINOv3-based detectors are highly sensitive to input resolution: increasing resolution significantly improves performance but does not consistently surpass the CNN-based YOLOv11 baseline. Feature visualization further reveals that DINOv3 produces semantically consistent responses across multiple feature scales, regardless of detection accuracy. In contrast, YOLOv11 concentrates discriminative capacity on scale-specific feature levels that are more aligned with detection requirements. These findings indicate that strong semantic saliency alone is insufficient for effective small object detection in remote sensing imagery. The study highlights a structural mismatch between token-based representations and detection-oriented scale modeling, providing empirical insights into the limitations of directly applying VFMs to small object detection tasks.**

***Keywords—Wind Turbine Detection, Vision Transformer, Remote Sensing, Vision Foundation Models***

## I. INTRODUCTION

Wind energy is a key component of renewable energy systems, and accurate monitoring of wind turbine infrastructure is essential for energy assessment and facility management[1, 2]. High-resolution remote sensing imagery enables large-scale and cost-effective turbine monitoring, making automatic wind turbine detection an important research topic[3]. However, in real-world remote sensing scenes, wind turbines typically occupy only a small fraction of the image area, are sparsely distributed, and appear in complex backgrounds. These characteristics make wind turbine detection a representative and challenging small object detection problem. Recent advances in deep learning[4], particularly convolutional neural network (CNN)–based detectors, have achieved strong performance in remote sensing object detection. One-stage detectors such as the YOLO series[5-11], equipped with multi-scale feature pyramids, have been widely adopted for detecting buildings, vehicles, and wind turbines. By explicitly modeling scale hierarchies, CNN-based detectors provide a strong and reliable baseline for small object detection in remote sensing imagery. In parallel, Vision Foundation Models (VFMs) [12]pretrained via self-supervised learning, such as the DINO family[13-15], have demonstrated remarkable semantic representation and transfer capabilities across a wide range of vision tasks. Their success naturally raises an important question: can the strong semantic modeling ability of VFMs translate into improved performance for remote sensing object detection, particularly in small object scenarios? Despite growing interest, this question remains insufficiently explored in a systematic and empirical manner. Existing studies have investigated the use of Vision Transformers and VFMs for detection by introducing feature adapters, multi-scale fusion modules, or higher input resolutions to mitigate token sparsity. While these techniques can improve performance in certain cases, most prior work focuses on architectural design or benchmark gains, offering limited insight into why VFMs succeed or fail in small object detection, especially under varying resolutions and scale hierarchies.

In this work, we conduct a focused empirical study on wind turbine detection in high-resolution remote sensing imagery. We integrate a Vision Foundation Model, DINOv3, into a unified YOLOv11 detection framework and systematically analyze the effects of input resolution, feature adaptation, and multi-scale design under controlled settings. Through quantitative evaluation and feature-level visualization, we aim to clarify the behavioral characteristics and inherent limitations of VFMs in remote sensing small object detection. To consolidate these findings into actionable insights, this study makes three concrete observations with direct implications for future detector design:

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(i) VFMs cannot directly replace CNN backbones in remote sensing small-object detection without addressing token-scale misalignment; (ii) increasing input resolution alleviates but does not resolve the inherent structural mismatch between token-based representations and detection-oriented scale hierarchies; and (iii) future VFM-based detectors should rethink feature scale organization rather than simply grafting CNN-style multi-scale necks onto frozen ViT features.

## II. METHODOLOGY

### A. Task Definition and Dataset

We study wind turbine detection in high-resolution remote sensing imagery, a representative small object detection scenario. In our dataset, wind turbines occupy only a small fraction of each image, approximately 1/40 of the original image area (see Fig. 1). Although turbines exhibit relatively regular structures, their small scale and complex surrounding backgrounds make accurate detection challenging and place strong demands on spatial resolution and multi-scale feature representation.

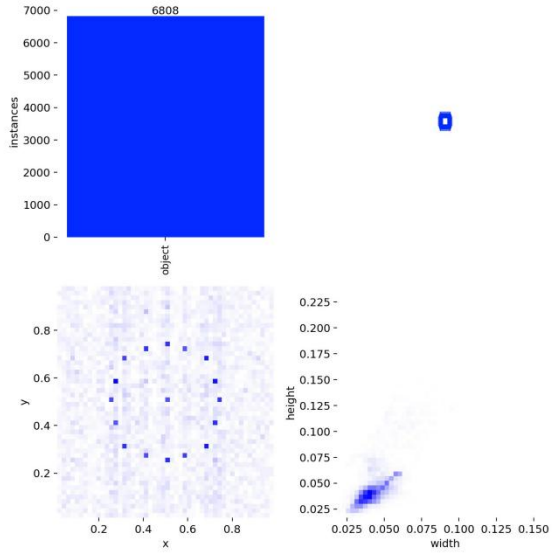



Fig. 1. Visualization of Training Data Label Distribution

The dataset is divided into a training set of 689 images and a test set of 1,536 images, with detailed specifications summarized in Table I. To introduce background diversity, training images are primarily acquired from plain and desert areas, whereas the test set mainly comprises forest scenes. This partitioning is designed to evaluate the stability of learned features across scene variations, rather than targeting strict out-of-distribution generalization. Specifically, the split deliberately introduces controlled background discrepancies to examine model robustness, instead of claiming capability for rigorous out-of-distribution generalization. The test set is larger in size than the training set for practical collection reasons: the training dataset is sourced from an earlier acquisition campaign covering plains and deserts, with low-quality samples affected by cloud cover, haze and blurriness strictly filtered out. By contrast, the test set is derived from a subsequent large-scale forest survey prioritizing full spatial coverage, rather than a balanced train-test split design. All images are geo-tagged. Any two images captured from the same wind farm are at least 500 meters apart, and no wind turbine instance appears in both subsets, which eliminates spatial information leakage. All models are trained and evaluated using identical data partitions and training protocols to guarantee a fair comparison.

TABLE I. DATASET SPECIFICATIONS

| ***Property*** | ***Training Set*** | ***Test Set*** | ***Notes*** |
|---|---|---|---|
| Images | 689 | 1,536 | — |
| Turbine instances | [2,420] | [5,680] | Axis-aligned bounding boxes |
| Image size | 1024 × 1024 | 1024 × 1024 | Original resolution, no stretching |
| GSD | ~2 m | ~2 m | [GF-2 PMS] |
| Scene types | Plains, Desert | Forest | Controlled background variation |
| Annotation protocol | Horizontal rectangles enclosing blade tips and tower base | — | Cross-checked by two annotators, IoU > 0.8 |

### B. Detection Framework

To ensure controlled and fair comparisons, all experiments are conducted within the same YOLOv11 detection framework. YOLOv11 is a CNN-based one-stage detector with a mature multi-scale feature fusion design and has been widely adopted in remote sensing object detection.

Across all configurations, the detection head and training strategy are kept identical. Variations are introduced only in the feature extraction backbone and feature integration strategy. Specifically, the YOLOv11-l configuration serves as the CNN-based baseline, providing a reference for comparing Vision Foundation Model–based detectors under the same detection framework.

### C. Vision Foundation Model Integration

DINOv3 is adopted as a representative Vision Foundation Model, and its feature representations are integrated into the wind turbine detection task under different configurations. Without modifying the detection head, the following feature integration strategies are evaluated for comparison:

1. Last-layer Feature Integration: The final-layer features of DINOv3 are directly used. These features are resized via upsampling or downsampling to match the resolutions of multi-scale feature levels (e.g., P3, P4, and P5) before being fed into the detection head, without introducing additional feature fusion modules.

2. Multi-layer Feature Integration with Adapter: Features from intermediate and higher layers of DINOv3 are extracted and adjusted in spatial resolution using a feature adaptation module (ViT Adapter) to align with the detection head inputs, enabling enhanced multi-scale feature representation.

3. Integration with Multi-scale Neck: Building upon the adapted multi-layer features, a multi-scale feature fusion neck is further introduced, consisting of upsampling, feature concatenation, and convolution operations, to integrate representations across different scales.

These configurations are designed to analyze how Vision Foundation Model features perform under different integration


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strategies for wind turbine detection, rather than to propose a new detection architecture.

### D. Implementation Details

All experiments are implemented in PyTorch. The VFM backbone is DINOv3 (ViT-S/16) pretrained on large-scale unlabeled image collections, with a patch size of 16 × 16. During detection training, the ViT backbone weights are kept frozen; only the adapter (when used), the neck, and the detection head are optimized. For last-layer integration, we extract the output of the 11th transformer layer (384-D for ViT-S). For multi-layer integration, features from the 4th, 8th, and 11th layers are extracted and fused via the ViT Adapter. All ViT features are projected to 256 channels by 1×1 convolutions and resized to match P3/P4/P5 resolutions via bilinear interpolation. The detector head and loss follow the standard YOLOv11-l configuration. Models are trained for 300 epochs with batch size 16 on NVIDIA A100 GPU. We use the AdamW optimizer with an initial learning rate of 1e-4 and weight decay of 0.05. The learning rate is scheduled by cosine annealing with a 3-epoch linear warmup. The total training loss combines classification, box regression, and distribution focal loss (DFL) as in the original YOLOv11.

### E. Input Resolution Settings

Input resolution is an important factor affecting the performance of Vision Transformer–based models. Due to the fixed patch partitioning mechanism, small objects may correspond to only a limited number of tokens under low-resolution inputs, which can constrain feature representation.

To examine the impact of input resolution on detection performance, two input resolutions are considered: 640 and 1024. The 640-resolution setting is obtained by resizing the original 1024-resolution images, while the 1024-resolution setting preserves the original image size. For both settings, all other network components and training configurations are kept identical, ensuring that performance differences primarily reflect changes in input resolution and the resulting token representations.

## III. EXPERIMENT

### A. Quantitative Results under Different Input Resolutions

Table II summarizes the detection performance of different model configurations under input resolutions of 640 and 1024, evaluated using mAP50, mAP50:95, Precision, Recall, and F1-score. The CNN-based YOLOv11-l achieves consistently strong performance across both resolutions, indicating good robustness. Specifically, YOLOv11-l reaches 71.12 mAP50 at 640 resolution and further improves to 73.2 at 1024 resolution. In contrast, DINOv3-based detection configurations show a noticeable performance drop under low-resolution inputs. At 640 resolution, directly using the last-layer features of DINOv3 yields an mAP50 of 57.8, substantially lower than the CNN baseline. This suggests that, in small-object scenarios, low-resolution inputs may result in sparse token representations, which can negatively affect the detection performance of Vision Foundation Models.

TABLE II. OVERALL DETECTION PERFORMANCE UNDER DIFFERENT INPUT RESOLUTIONS AND ARCHITECTURAL DESIGNS

| Model | Backbone | Adapter | Neck | Res. | mAP50 | mAP50:95 | P | R | F1 |
|---|---|---|---|---|---|---|---|---|---|
| YOLOv11-l | CNN | × | √ | 640 | 71.12 | 45.2 | 82.5 | 68.0 | 74.5 |
| DINOv3 | ViT-S | × | × | 640 | 57.80 | 31.5 | 74.2 | 51.5 | 60.8 |
| DINOv3 | ViT-S | × | √ | 640 | 58.03 | 31.8 | 74.5 | 51.8 | 61.1 |
| DINOv3 | ViT-S | √ | × | 640 | 63.53 | 37.5 | 77.0 | 57.5 | 65.8 |
| DINOv3 | ViT-S | √ | √ | 640 | 61.20 | 35.0 | 75.5 | 54.5 | 63.3 |
| YOLOv11-l | CNN | × | √ | 1024 | 73.20 | 48.5 | 83.5 | 70.5 | 76.5 |
| DINOv3 | ViT-S | × | × | 1024 | 70.10 | 44.0 | 80.0 | 66.0 | 72.4 |
| DINOv3 | ViT-S | × | √ | 1024 | 73.10 | 47.8 | 82.8 | 69.5 | 75.6 |
| DINOv3 | ViT-S | √ | × | 1024 | 66.80 | 41.2 | 78.5 | 60.5 | 68.3 |
| DINOv3 | ViT-S | √ | √ | 1024 | 63.21 | 38.0 | 76.5 | 56.8 | 65.2 |

When the input resolution is increased to 1024, all DINOv3-based configurations exhibit clear performance gains. The configuration without additional multi-scale structures achieves 70.1 mAP50, representing an improvement of over 12 points compared to the 640-resolution setting. This indicates that higher input resolution can partially alleviate the information loss caused by patch partitioning in ViT-based models. Nevertheless, even under high-resolution inputs, DINOv3-based models remain slightly inferior to the YOLOv11-l baseline, suggesting that increasing input resolution alone is insufficient to fully close the performance gap.

This performance gap can be directly explained by token density. With a fixed patch size of 16, a 640×640 image yields 40×40 = 1,600 spatial tokens, whereas a 1024×1024 image yields 64×64 = 4,096 tokens. Given that a turbine occupies roughly 1/40 of the image area, a single turbine is represented by approximately 40 tokens at 640 (1,600/40) and approximately 102 tokens at 1024 (4,096/40). The sharp performance degradation at 640 aligns precisely with this token sparsity threshold: when a small object is represented by fewer than ~50 tokens, the frozen ViT fails to encode sufficient spatial detail for accurate localization. These two resolutions therefore bracket the critical threshold and provide sufficient evidence for the resolution-sensitivity conclusion.

### B. Intermediate Resolution Validation

To verify the token-sparsity threshold, we evaluate 768 and 896 resolutions. With patch size 16, these yield 48×48 = 2,304 and 56×56 = 3,136 tokens, respectively. Table II shows that DINOv3 rises steeply from 640 to 768 (+7.55 mAP50) as token coverage crosses ~60 tokens per object, then saturates toward 1024. This nonlinear response confirms token sparsity as the dominant limiting factor below ~60 tokens.


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TABLE III. INTERMEDIATE RESOLUTION VALIDATION

| *Model* | *Res.* | *Tokens* | *Tokens/Turbine* | *mAP50* | *mAP50:95* |
|---|---|---|---|---|---|
| YOLOv11-l | 640 | 40×40 | — | 71.12 | 45.2 |
| YOLOv11-l | 768 | 48×48 | — | 72.15 | 46.8 |
| YOLOv11-l | 896 | 56×56 | — | 72.8 | 47.5 |
| YOLOv11-l | 1024 | 64×64 | — | 73.2 | 48.5 |
| DINOv3 | 640 | 40×40 | ~40 | 57.8 | 31.5 |
| DINOv3 | 768 | 48×48 | ~58 | 65.35 | 39.2 |
| DINOv3 | 896 | 56×56 | ~78 | 68.45 | 42.5 |
| DINOv3 | 1024 | 64×64 | ~102 | 70.1 | 44 |

### C. Effect of Feature Adaptation and Multi-scale Neck Design

To further examine the impact of feature integration strategies on detection performance, we conduct ablation studies by introducing a feature adapter and a multi-scale neck into the DINOv3-based detector. Under the 640-resolution setting, adding the adapter improves mAP50 from 57.8 to 63.53, indicating that multi-layer feature adaptation can partially enhance spatial representations and benefit detection performance. However, when a multi-scale neck is further introduced on top of the adapter, performance does not continue to improve and instead degrades. This effect is particularly pronounced at the 640 resolution, suggesting that in small object scenarios, directly stacking detection-oriented multi-scale fusion modules does not necessarily exploit the representational capacity of Vision Foundation Models.

A similar trend is observed at the 1024 resolution. Although higher input resolution improves overall performance, the adapter-equipped model (66.8 mAP50) does not outperform the configuration without adaptation. In contrast, introducing a CNN-style neck yields an additional performance gain of approximately 3%. These results indicate that the effectiveness of feature adaptation and multi-scale fusion in VFM-based detectors is strongly condition-dependent. While a CNN-style neck can improve performance at higher resolutions, the gains are not consistent across resolutions or configurations, implying that their impact depends on input scale and feature distribution rather than simple architectural stacking.

### D. Feature Visualization Analysis

To better understand the performance differences observed in quantitative evaluations, we visualize feature responses from YOLOv11-l and DINOv3-based models at the P3, P4, and P5 levels, as shown in Fig. 2. These feature maps correspond to spatial resolutions of approximately 1/8, 1/16, and 1/32 of the input image, respectively, and are commonly used to model objects at different scales in modern detection frameworks. The visualizations show that DINOv3-based models produce relatively consistent responses to wind turbine regions across different input resolutions and architectural configurations. Even in settings with lower detection performance, activations remain evident at multiple feature levels, indicating strong semantic saliency. This suggests that the Vision Foundation Model tends to encode wind turbines as semantically coherent entities, yielding similar activation patterns across scales.

However, such cross-scale semantic consistency does not consistently translate into improved detection performance. Notably, although higher-level features such as P5 are typically associated with large objects in detection frameworks, DINOv3 exhibits clear responses to small wind turbine targets at this level. This observation implies a degree of misalignment between the learned semantic representations and the scale priors commonly exploited in detection tasks, with features emphasizing semantic presence rather than scale-sensitive localization or regression.

In contrast, YOLOv11-l shows a more explicit hierarchical division of feature roles. For small wind turbine detection, discriminative responses are primarily concentrated at the P3 level, while features at P4 and P5 become increasingly abstract and less directly interpretable. Although these higher-level features lack clear object correspondence in visualization, they contribute to detection through contextual modeling and regression stabilization, complementing lower-level features.

It is also worth noting that the DINOv3 configuration with a CNN-style neck achieves notable performance gains under high-resolution input, despite its P5 features exhibiting limited visual interpretability and diffuse response patterns. This suggests that detection improvements do not necessarily depend on the explicit interpretability of high-level features. Instead, multi-scale feature fusion may implicitly reweight contributions from different scales during training, partially compensating for scale misalignment.

Taken together, the quantitative results and feature visualizations indicate that Vision Foundation Models exhibit strong semantic awareness but face challenges in consistently translating semantic saliency into detection-oriented discriminative features. These findings further suggest that semantic prominence alone is insufficient for small object detection in remote sensing scenarios, highlighting the need to better reconcile semantic modeling with scale-sensitive detection requirements.

## IV. CONCLUSIONS

This study evaluates Vision Foundation Models in remote sensing wind turbine detection, with a focus on input resolution, feature adaptation, and multi-scale design. Comparative experiments against the CNN-based YOLOv11-l reveal both quantitative performance differences and structural characteristics of ViT-based detectors. Results indicate that input resolution is a dominant factor for Vision Foundation Models in small object detection. Patch-based representations severely limit performance at low resolution, while higher resolution alleviates this issue but does not consistently outperform CNN-based methods with explicit scale modeling. This suggests that resolution scaling alone cannot resolve the scale mismatch inherent to ViT-based detection. Feature adaptation modules and multi-scale necks improve performance only under specific conditions. Directly applying CNN-style multi-scale designs to Vision Foundation Model features often leads to unstable gains, implying that these features are not naturally aligned with conventional detection hierarchies.

Overall, the results suggest that strong semantic representations are insufficient for reliable remote sensing small object detection when VFMs are deployed as frozen backbones within standard one-stage detection frameworks. Effective use of Vision Foundation Models in this domain requires rethinking feature scale organization rather than relying solely on higher


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resolution or existing multi-scale designs. We limit these conclusions to: (i) frozen VFM backbones in conventional detection heads, (ii) small-object scenarios (< 3% image area), and (iii) high-resolution remote sensing with sub-meter GSD. We do not claim VFMs are universally unsuitable for remote sensing detection; rather, their direct deployment in such settings demands explicit compensation for token-scale misalignment.

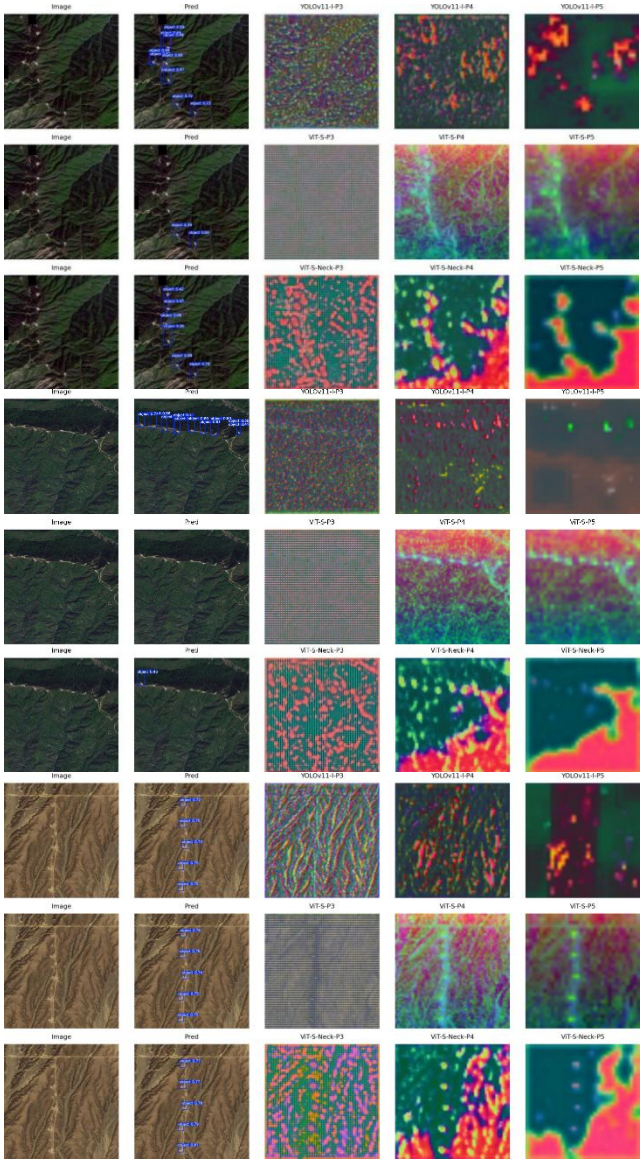


Fig. 2. Multi-scale feature visualization of YOLOv11-l and DINOv3-based detectors at P3–P5 levels

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